ANAND NAGARAJ

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PROFILE SUMMARY

Dedicated Master's student in Design and Test of Integrated Circuits at TU Chemnitz, combining 3+ years of industry design experience with disciplined academic research. Proven track record in pipelined RISC-V microarchitectures, open-source RTL-to-GDSII ASIC implementation, and formal/UVM verification frameworks. Committed to contributing high-quality digital architectures to cutting-edge research environments.

EDUCATION & ACADEMIC QUALIFICATIONS

Technische Universität Chemnitz **Chemnitz, Germany**
Master of Science in Design and Test of Integrated Circuits (DTIC) *Apr 2026 – Present*

- **Completed Coursework (Semester 1):** Verification of Digital Systems, Integrated Circuit Design (Transistor Level), Design of Heterogeneous Systems, Advanced Integrated Circuit Technology, Technologies of Micro/Nano Systems.
- Completed all laboratory components and practical chip verification assignments.

Visvesvaraya Technological University (VTU / BMSIT) **Bengaluru, India**
Bachelor of Engineering in Electronics & Telecommunication Engineering *Aug 2018 – Apr 2022*

- **Academic Standing:** Graduated with an equivalent German GPA of 1.8 (CGPA 8.51 / 10.0).
- **Honors:** Conferred **Best Outgoing Student** in the Department; Recipient of Karnataka State Merit Scholarship.
- **Key Modules:** CMOS VLSI Design, Computer Architecture, Digital System Design, Microprocessors.

Maven Silicon **Bengaluru, India**
Advanced VLSI Design and Verification (6-Month Full-Time Vocational Program) *Apr 2022 – Sep 2022*

- Practical training in SystemVerilog, modular UVM testbench architectures, AMBA protocols (AXI/AHB/APB), and QuestaSim.

PROFESSIONAL WORK EXPERIENCE

IndusIP Labs Pvt. Ltd. **Bengaluru, India**
VLSI Design & Verification Engineer (Founding Team) *Nov 2024 – Feb 2026*

- Worked extensively with the open-source **RISC-V Ibex core**, implementing custom instruction extensions and dedicated hardware execution logic for specialized workloads, validated via FPGA prototyping.
- Designed synthesizable SystemVerilog RTL for **2D-mesh Network-on-Chip (NoC) routers**, implementing virtual channel allocators, round-robin arbiters, and crossbars.
- Implemented **SystemVerilog Assertions (SVA)** across asynchronous FIFO clock-domain crossings (CDC) to eliminate transaction deadlocks under heavy synthetic traffic.
- Built simulation testbenches in SystemVerilog and Cocotb; developed automated Python regression frameworks to aggregate simulation logs and track functional coverage.

Sinergia Media Labs (Simelabs) **Kochi / Bengaluru, India**
RTL Design & Verification Engineer *Sep 2022 – Oct 2024*

- Designed synthesizable Verilog RTL for high-throughput **PCIe Transaction Layer IP**, implementing header parsing, packet framing, and flow control buffer management.
- Implemented end-to-end data integrity protection using 32-bit CRC (ECRC) generation/checking and poisoned packet detection logic.
- Developed modular **UVM testbenches** and Python Cocotb co-simulation harnesses to validate transaction ordering and protocol corner cases.
- Synthesized RTL blocks with Synopsys Design Compiler, verifying setup/hold margins and static timing constraints.

KEY SCIENTIFIC & ENGINEERING PROJECTS

Moola-V (MR5): 5-Stage Pipelined RV32IM Processor Core *SystemVerilog / Cocotb*

- Architected an in-order, 5-stage pipelined RV32IM core in synthesizable SystemVerilog with full RAW hazard forwarding (EX-to-EX, MEM-to-EX) and cycle-accurate load-use stalls.
- Integrated dedicated datapath units including a single-cycle multiplier and a 32-cycle iterative radix-2 restoring divider.
- Verified 100% architectural compliance across all 49 RISCOF test suites co-simulated against Spike ISS via Python Cocotb.

Parameterized Dual-Port Synchronous SRAM Macro *SkyWater 130nm / OpenLane 2*

- Designed and verified a parameterized 16×8 dual-port synchronous SRAM macro module featuring independent write-masking and concurrent read/write arbitration logic within the open-source Tiny Tapeout framework.
- Executed the complete physical implementation flow from RTL down to GDSII using OpenLane 2 and OpenROAD, achieving clean DRC/LVS verification and static timing closure.

Mixed-Signal Frequency Measurement & Interface System *TU Chemnitz Academic Project*

- Architected a dual-clock digital measurement unit in VHDL with cooperating FSMs and request-acknowledge CDC handshakes.
- Modeled a continuous-time sensor front-end in VHDL-AMS with digital threshold detection logic and a UART transmitter.

AHB-to-APB Protocol Bridge & Verification Suite *Verilog / SystemVerilog*

- Designed synthesizable RTL for an AMBA-compliant AHB-to-APB bridge, implementing cycle-accurate FSM control logic to translate pipelined AHB transfers into multi-phase APB bus cycles.
- Implemented address decoding, wait-state propagation via `HREADYOUT`, error response mapping, and transient register buffering.
- Developed a self-checking SystemVerilog testbench with assertions (SVA) and functional coverage to verify burst sequences and transfer timeouts.

PUBLICATIONS

- **A. Nagaraj et al.**, "*Peripheral Shared Memory Over PCI Express Link*," in **Proceedings of the 7th IEEE International Conference on Innovative Data Communication Technologies and Application (ICIDCA 2025)**. Analyzed shared-memory throughput, buffer sizing trade-offs, and transaction latency across high-speed PCIe interfaces.

HONORS, AWARDS & SELECTED PARTICIPATIONS

- **SEMICON Europa × TUM Venture Labs Hackathon (NextSEMI 2026):** Selected as one of only 25 participants across Europe for an intensive semiconductor prototyping and hardware sprint in Munich (Nov 2026).
- **State Merit Scholarship (Karnataka Board of Education):** Awarded 100% undergraduate tuition waiver for ranking among the top candidates in the state competitive examination.
- **Best Outgoing Student Award (2022):** Conferred by the Department of Electronics & Telecommunication Engineering, VTU/BMSIT, for sustained academic and leadership excellence.
- **District Kabaddi Championship:** Secured 2nd place in a competitive Under-19 District Tournament.

TECHNICAL & LANGUAGE SKILLS

- **Hardware Description & Modeling:** SystemVerilog (IEEE 1800), Verilog, VHDL, VHDL-AMS (Analog/Mixed-Signal Behavioral Modeling).
- **Architectures & Protocols:** AMBA (AXI4, AHB, APB), PCIe Gen3/4 Transaction Layer, Network-on-Chip (NoC), RISC-V (RV32IM, Ibex), SRAM.
- **Verification & Methodologies:** UVM, Cocotb (Python), SystemVerilog Assertions (SVA), Functional Coverage, CDC Analysis, RISCOF.
- **EDA Tools & Simulators:** QuestaSim, ModelSim, Cadence Xcelium, Synopsys VCS, Synopsys Design Compiler, GTKWave, Verilator.
- **Physical Design & Open Silicon Flows:** OpenLane 2, OpenROAD, Tiny Tapeout Flow, SkyWater 130nm PDK, KLayout, Magic.
- **Scripting & Automation:** Python, Bash/Shell, Linux/Unix, Git.
- **Languages:** English (Fluent, C1); German (A2 certified, enrolled in B1 at TU Chemnitz); Kannada (Native); Hindi.

COMMUNITY ENGAGEMENT & EXTRACURRICULAR ACTIVITIES

- **IEEE Youth Wing Chemnitz:** Student volunteer supporting technical events and student networking.
- **SMD Chemnitz:** Active participant in intercultural community sessions and campus integration initiatives.
- **Academic & Civic Coordination:** Coordinated departmental cultural and technical symposiums during undergraduate studies; volunteered during regional elections assisting elderly voters with polling administration.